

Examining Seattle Weather with Time Series Analysis: Predicting Wind Speed from Max Temperature using ARIMAX Models

Nick Oldfather

Introduction:

For my final project, I used a Kaggle dataset that has Seattle weather data from January 1st, 2012, to December 31st, 2015. The Kaggle poster collected the data from the National Oceanic and Atmospheric Administration (NOAA), which has a wealth of historical weather data.

The dataset uses 1-day timesteps and has no missing dates, so this includes 1,461 time points over these four years. The dataset has six fields: the date, precipitation in inches, maximum temperature in Celsius, minimum temperature in Celsius, wind speed in miles per hour, and the type of weather for the day, e.g., rain, sun, snow, etc. This was a dataset that I've used before in a machine learning course, where I used a variety of predictors to predict the weather type in Seattle. I wanted to see if this dataset could be used for a time series analysis, particularly, I wanted to examine if I could predict wind speed from max temperature, and what would be the best time series model to predict wind speed using these two features.

There were three hypotheses I wanted to test in my analysis. 1) Both max temperature and wind speed exhibit strong seasonal trends. 2) There is an inverse relationship between max temperature and wind speed. 3) Past values of max temperature and wind speed significantly predict current wind speed.

Methods:

To make my analyses more interpretable for American readers, I converted the max temperature from Celsius to Fahrenheit; assume all plots with temperature are Fahrenheit and plots with wind speed are in miles per hour (mph). Additionally, any time I refer to temperature in the following analysis, I am referring to the maximum temperature, not the minimum temperature, unless I explicitly state otherwise.

Figure 1 shows the distribution of max temperature and wind speed. Max temperature is approximately normally distributed, centered around 60 degrees Fahrenheit. The minimum is 29.12 degrees Fahrenheit, which initially seems quite high; however, when looking at the minimum temperature, it is -7.1 degrees, which seems quite reasonable. The maximum temperature is 96.08 degrees. Upon research, these temperature ranges also seem reasonable for Seattle weather patterns. The distribution of wind speed is skewed right, centered around 3 mph. The minimum wind speed is 0.4mph, and the maximum is 9.5mph.

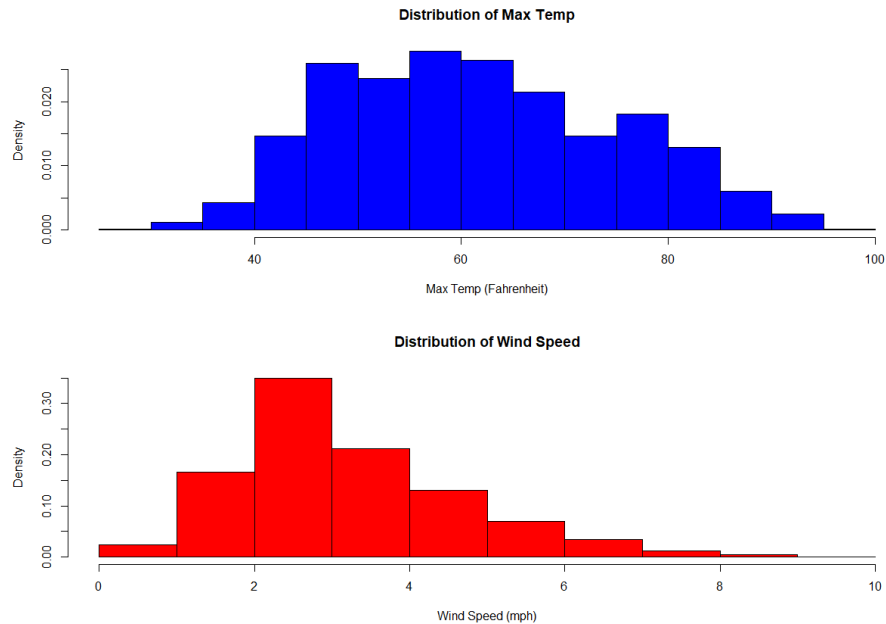


Figure 1: Distribution of Max Temperature and Wind Speed

Figure 2 shows a time series of both max temperature and wind speed over time. One thing that jumps out immediately is that there are strong cyclical trends for max temperature. The temperature is lower in the winter times, progressively gets warmer as summer approaches, and then progressively gets colder as winter approaches again. The wind speed time series is much noisier, and it is hard to tell if there is seasonality. It looks like it may be higher during the winter and colder during the summer, but it is not obvious.

Figure 3 shows both time series using a moving average filter of 31 days. This means each point is the average of 15 days before, the point itself, and 15 days after. The first 15 points in the time series and 15 points at the end are not included, as they can't be easily smoothed. The seasonal trend for max temperature is still obvious, and it seems we may be correct that there is a weak seasonal trend for wind speed. It does seem slightly higher in the winter months and slightly lower in the summer months.

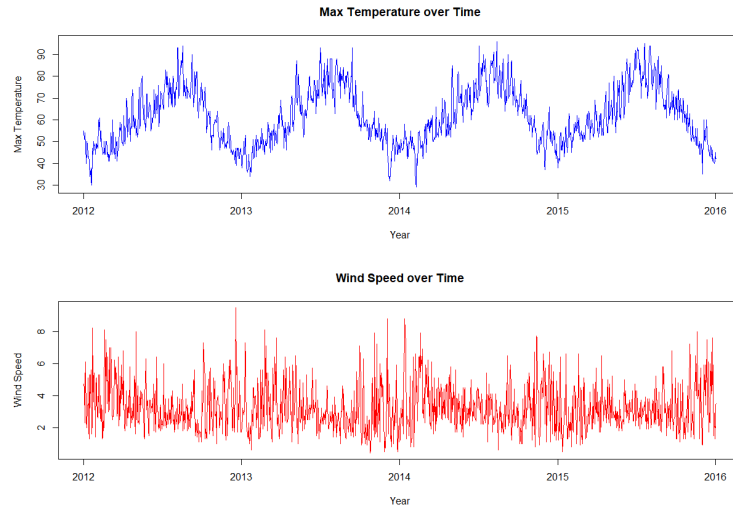


Figure 2: Time Series of Max Temperature and Wind Speed over Time

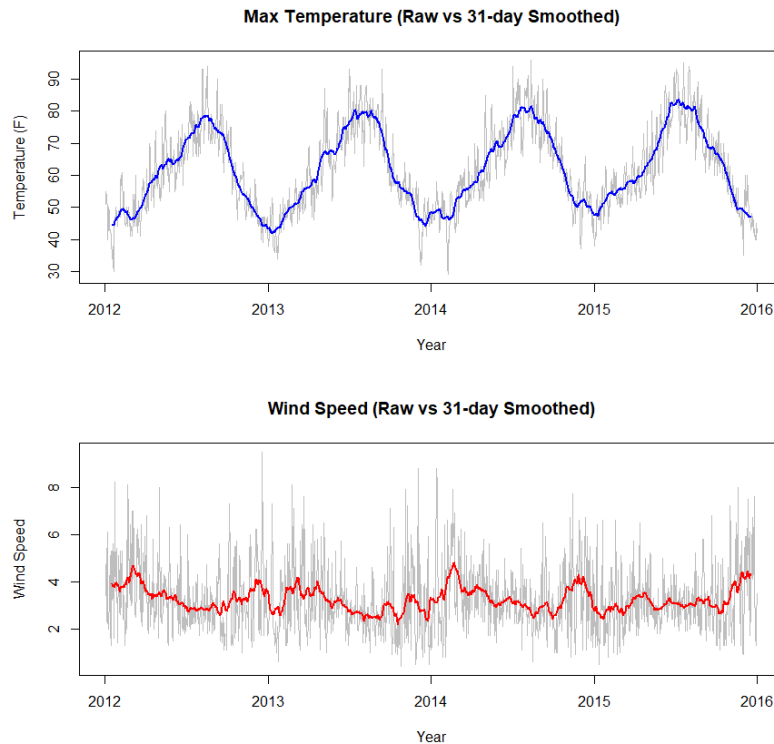


Figure 3: Time Series of Max Temperature and Wind Speed Smoothed using a 31-day Window

Figure 4 shows the periodogram for temperature and wind speed. There is a big spike at approximately $1/365$ for temperature, indicating that there is a big yearly cycle. Wind speed is much more spread out; however, there is a big spike at the beginning, indicating that there may be a weaker cycle. These periodograms are consistent with our evaluations from the time series plots.

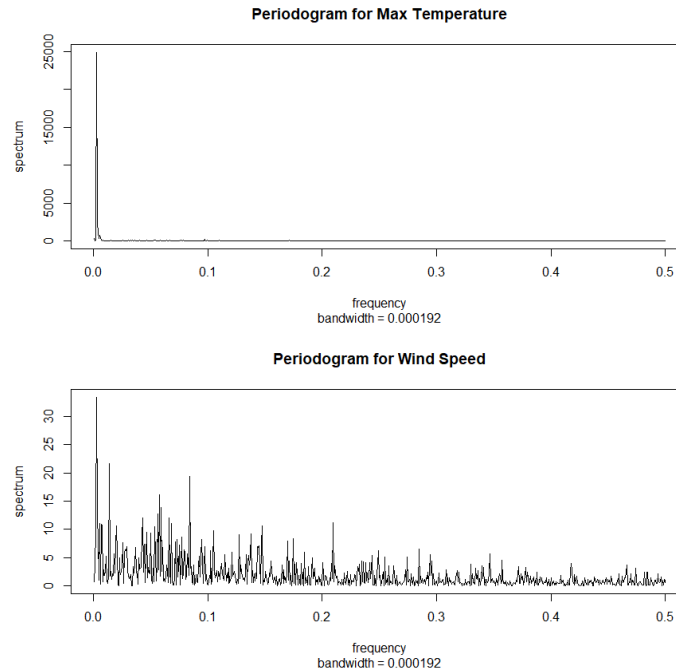


Figure 4: Periodogram of Max Temperature and Wind Speed

To properly evaluate the relationship between temperature and wind speed, as well as the best models to predict wind speed without seasonality, we need to detrend the data. To do this, I fit two linear models, with the same predictors, to predict temperature and wind speed. The predictors were four sinusoidal functions, two were using sine and two using cosine. All four use the variable date, which is the day of the year in chronological order. For example, January 1st, 2012 has date 1 and December 31st, 2015 has date 1461. The predictors were $\sin(2/365 \cdot \pi \cdot \text{date})$, $\cos(2/365 \cdot \pi \cdot \text{date})$, $\sin(4/365 \cdot \pi \cdot \text{date})$, $\cos(4/365 \cdot \pi \cdot \text{date})$. After fitting the linear regression models, I calculated the residuals, which I would use as data that had been “detrended”, meaning the seasonality had been removed.

A time series of the residuals for temperature and wind speed can be seen in Figure 5. The detrending appears to be successful, and the seasonality has been removed. This is especially obvious with the temperature plot.

Figure 6 shows the periodograms for temperature and wind speed after detrending. The massive spike at the beginning of the temperature plot has been removed, so the large seasonal cycle has been removed. The wind speed plot is also much more spread out, without the spike at the beginning, so the weak seasonal cycle seems to be at least partially removed from wind speed, too.

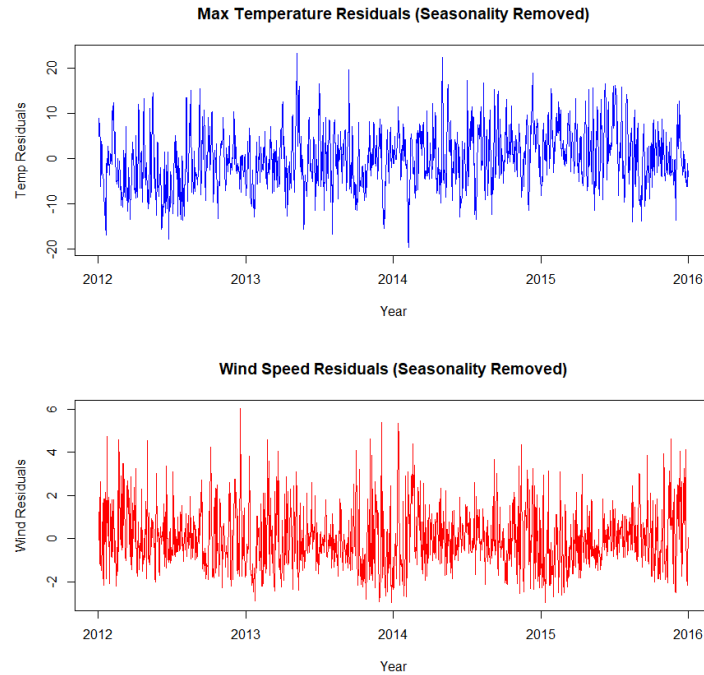


Figure 5: Time Series of Max Temperature and Wind Speed after Detrending

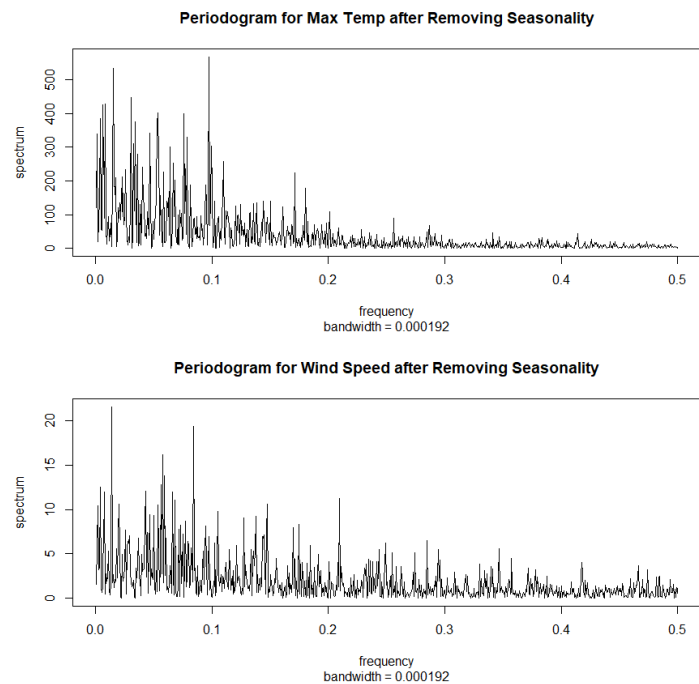


Figure 6: Periodogram for Max Temperature and Wind Speed after Detrending

Figure 7 shows the autocorrelation function (ACF) for temperature and wind speed, both before and after detrending. Once again, before detrending, there is a very strong cyclical trend for temperature and a weaker cyclical trend for wind speed, indicated by the oscillations over the

year. After detrending, the max temperature has significant autocorrelations for around 40 lags. Meaning that temperature seems to persist many days later; hot days are usually followed by hot days, and colder days are usually followed by cold days. An autoregressive process seems like a good model for temperature. Wind speed after detrending still has some oscillations, which could mean there could be a little persisting seasonality that we failed to remove after detrending. Additionally, wind speed seems to persist much shorter, only having significant autocorrelations for three lags; it does not abruptly stop, so it is likely also modeled using an autoregressive process, but a lower order process than temperature, which has longer significant lags.

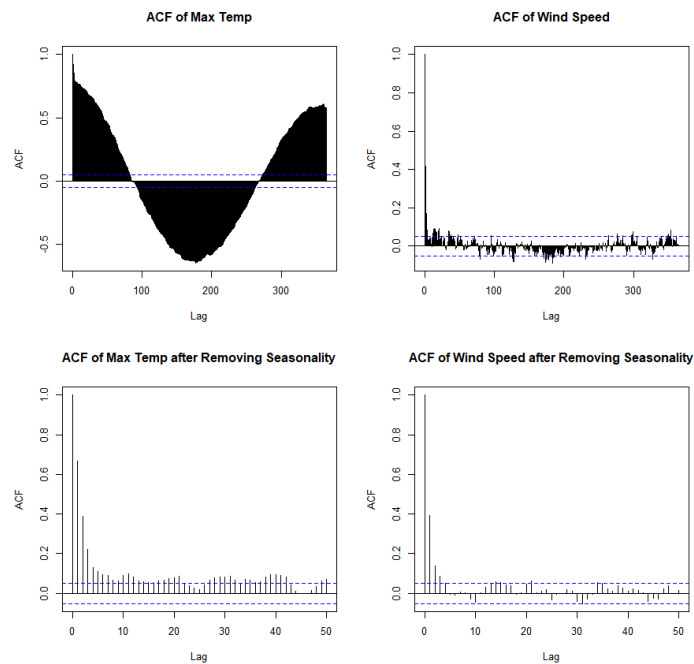


Figure 7: Autocorrelation (ACF) for Max Temperature and Wind Speed before and after Detrending

Figure 8 shows the cross-correlation function (CCF) for wind speed and max temperature before and after detrending. Before detrending, there were really strong seasonal trends, with wind speed and max temperature moving in opposite directions, within around 50 lags. This would then change to positive after 100 lags and back to negative after 200 lags, likely because of the seasons changing. After detrending, the relationship is much weaker. Most of the remaining statistically significant lags have a negative correlation. This indicates that wind speed and max temperature do seem to have an inverse relationship independent of seasonality; however, this relationship is much weaker once seasonality has been removed. Most of the negative correlation lags are on the right of the x-axis, indicating that wind speed may be following temperature and moving in the opposite direction.

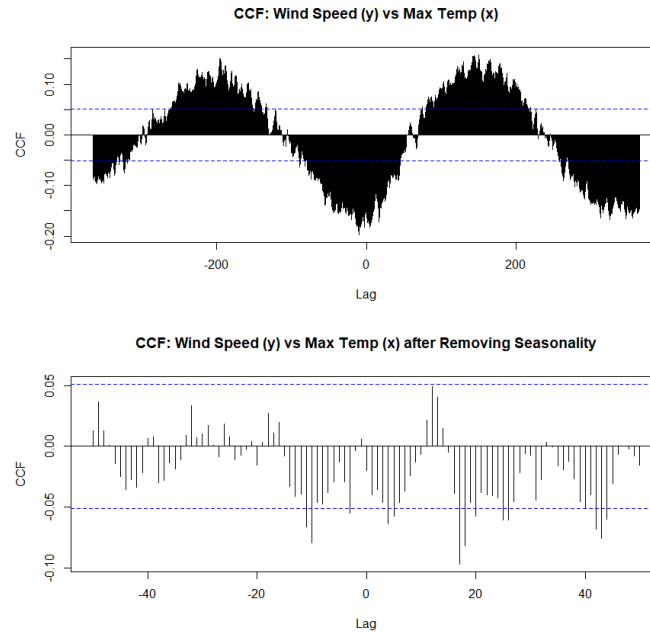


Figure 8: Cross Correlation (CCF) between Wind Speed and Max Temperature before and after Detrending

Figure 9 shows a spectral density analysis of max temperature and wind speed before and after detrending, using a window size of 31 days. The spectral density plots from before detrending are very similar to their periodograms, with a big seasonal spike for temperature and a smaller seasonal spike for wind speed. After detrending, as previously mentioned with their periodograms, the big seasonal spike at $1/365$ has been removed for temperature, and the wind speed plot is much more spread out. We've done a good job removing seasonality with our detrending method.

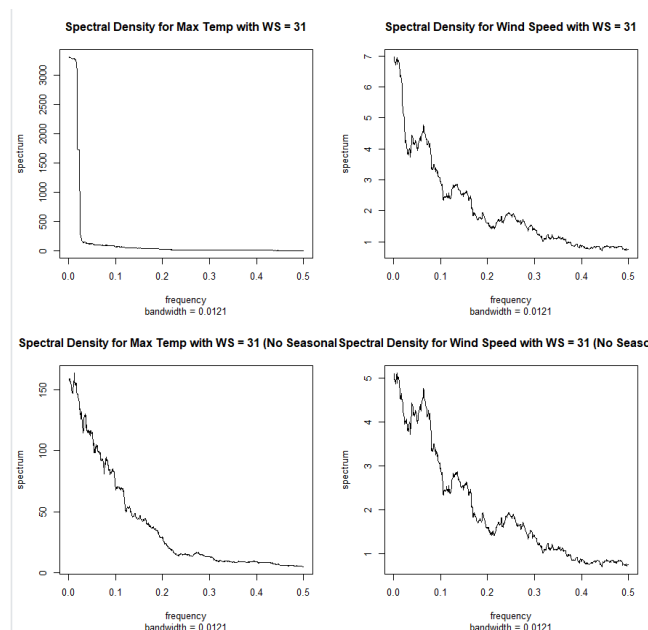


Figure 9: Spectral Density Analysis of Max Temperature and Wind Speed before and after Detrending using a Window Size of 31 Days

Results:

To predict current-day wind speed, I tested a variety of ARIMA and ARIMAX models and compared their AICs and BICs to find the best model with the fewest parameters. I also compared them to a baseline linear regression model, which I called model0, that simply predicts wind speed using current temperature.

Table 1 shows the models I tested, along with their AICs and BICs. All models, except for the baseline model, used ARIMA/ARIMAX (1, 0, 0) models, first-order autoregressive models. These parameters (p, d, q) were systematically selected by the auto.arima function from the forecast package as the best parameters as they minimized the model's BIC when using the given predictors.

Model #	Model Parameters	AIC	BIC
0	lm(wind ~ temp)	5165.412	5181.269
1	ARIMA(wind ~ wind lag 1)	4926.495	4942.351
2	ARIMAX(wind ~ wind lag 1 + temp)	4913.721	4934.863
3	ARIMAX(wind ~ wind lag 1 + temp lag 1)	4919.153	4940.295
4	ARIMAX(wind ~ wind lag 1 + temp + temp lag 1)	4915.695	4942.122
5	ARIMAX(wind ~ wind lag 1 + temp + temp lag 1 + temp lag 2)	4917.396	4949.109

Table 1: AIC and BIC for Tested Models.

All of these models were evaluated using every date in our dataset, except the first two: January 1st, 2012, and January 2nd, 2012, as the first two dates don't have two prior days to lag from for model 5 to use for its predictors, and we wanted to ensure AIC and BIC were calculated on the same data for equivalent comparison.

All ARIMA and ARIMAX models had significantly lower AIC and BIC than the baseline linear regression model and seem to perform significantly better. Model 2, the ARIMAX model that uses wind speed lag 1 and temperature had the lowest AIC and BIC. Several of our models are nested, so we used a chi-squared test to check that the extra predictors in the more complicated models were adding value. After comparing models 1 and 2, we get a p-value of approximately 0.0001, indicating that adding current temperature significantly improves our model. After comparing models 2 and 4, we get a p-value of 0.8709, indicating that adding temp lag 1 doesn't significantly improve our model. Model 2 has the lowest AIC and BIC, all of its predictors significantly improve the model, and adding additional predictors doesn't improve it; it is our best

model for predicting current wind speed. Additional lags for wind speed and temperature are unnecessary.

Figure 10 shows the residuals for our best model, model 2. Overall, the residuals look good. The spectral analysis plot, which uses a window size of 31, looks fairly flat, and there doesn't seem to be any big cycles. The time series plot seems fairly random, and in the ACF plot, only one lag is weakly statistically significant, which is expected for thirty lags when using a p-value of 0.95. Additionally, a Ljung-Box test has a p-value of 0.3948. This is much larger than 0.05. There is no evidence of leftover autocorrelation in the residuals. Overall, our residuals look like white noise. Our model appears to have done a good job of capturing the systematic trends in wind speed.

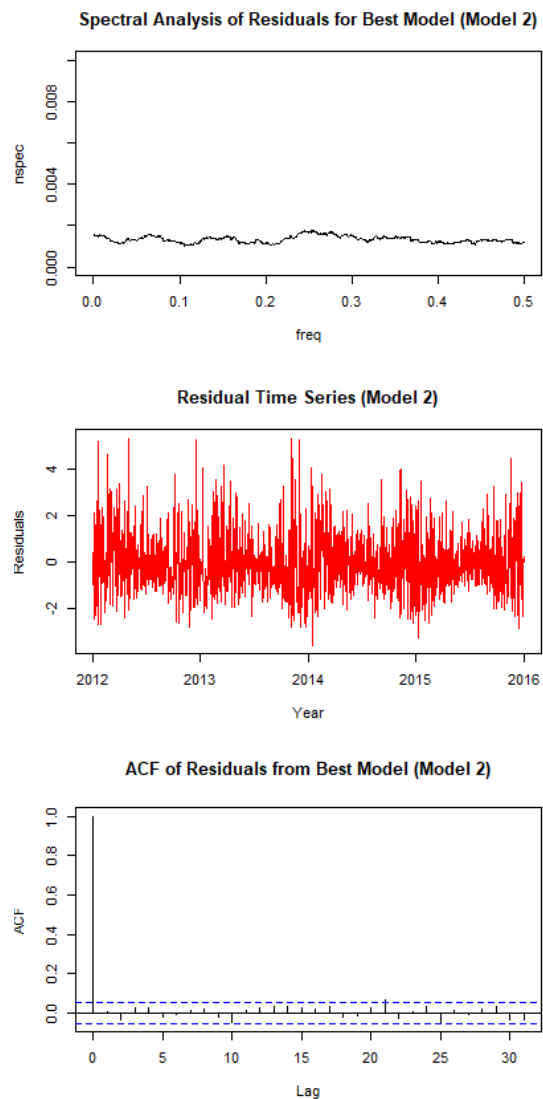


Figure 10: Residuals for our Best ARIMAX Model (Model 2)

Table 2 shows the coefficients for the parameters for our best model, model 2, and their respective p-values. All of the parameters are statistically significant, evidenced by the low p-values. This supports our previous chi-square analysis findings that all model 2 parameters are necessary. Wind speed preserves a lot of its information from the previous day, with a coefficient of 0.4. There is an inverse relationship between temperature and wind speed for the day. For each degree temperature increase, the wind speed drops by 0.028 miles per hour.

Parameter	Coefficient	P-value
Wind lag 1	0.400	< 0.001
Intercept	3.698	< 0.001
Temp	-0.028	< 0.001

Table 2: Coefficients and P-values for Parameters in our Best ARIMAX Model (Model 2)

We also created a Vector Auto Regressive model (VAR) to see if both temperature and wind speed are impacting each other. We used VARselect from the vars package to select the number of lags with the lowest BIC given our parameters, which ended up being lag 1. Our VAR models had the following equations: wind = wind lag 1 + temp lag 1 + intercept, and temp = wind lag 1 + temp lag 1 + intercept. The coefficients and p-values for these parameters can be seen in Table 3.

Equation Predicting	Parameter	Coefficient	P-value
wind	Wind lag 1	0.403	< 0.001
wind	Temp lag 1	-0.017	< 0.001
wind	Intercept	2.215	< 0.001
temp	Wind lag 1	-0.110	0.035
temp	Temp lag 1	0.920	< 0.001
temp	Intercept	1.669	< 0.001

Table 3: Coefficients and P-values for Parameters in VAR Model Equations.

We compared the VAR model equation for wind with our best ARIMAX model for wind. The VAR model had an RMSE of 1.301 while the ARIMAX model had an RMSE of 1.300. The ARIMAX model seems to perform either about the same or slightly better.

We also tested if temperature Granger-causes wind speed and if wind speed Granger-causes temperature. This is testing if one variable's predictions using both variable y and variable x are

statistically significant beyond just using values of variable x for prediction. Essentially, it checks if using the extra variable improves predictions. Temperature does Granger-cause wind speed with a p-value of 0.0003. Wind speed also Granger-causes temperature with a p-value of 0.0347. There is bidirectional Granger-causality between the two variables. Temperature changes seem to impact wind speed, and wind speed changes seem to impact temperature.

Discussion:

From the previous analysis, we can make the following conclusions about our initial hypotheses:

- 1) I was partially correct; the max temperature does exhibit strong seasonality. Wind speed exhibits weak seasonality. This can be seen in the time series and spectral analysis plots before detrending.
- 2) At face value, I was correct; max temperature and wind speed have a strong inverse relationship. However, this relationship is so strong due to seasonality, and once the data has been detrended, the relationship is much weaker. After detrending, there is still a weak inverse relationship. This can be seen when looking at the detrended CCF and the negative coefficient value for temperature when predicting wind speed in our best model.
- 3) I was incorrect. Wind speed lag 1 did end up being a significant predictor in our best model, but past values of temperature were not statistically significant for predicting wind speed, only current values of temperature. No values of wind speed or temperature were statistically significant beyond lag 1. Our best model for wind speed only had wind speed lag 1 and current temperature as predictors.

Beyond our initial hypotheses, we also found that the VAR model that has the lowest BIC with respect to our predictors only used 1 lag and had approximately equal or slightly worse performance than our best ARIMAX model. Additionally, wind speed Granger-causes temperature, and temperature Granger-causes wind speed. There is bidirectional Granger-causality between the two variables.

Works Cited:

Awasthi, A. (2023, August 18). *Seattle Weather Prediction Dataset*. Kaggle.
<https://www.kaggle.com/datasets/petalmc/seattle-weather-prediction-dataset>

National Centers for Environmental Information. National Centers for Environmental Information (NCEI). (n.d.). <https://www.ncei.noaa.gov/>